

MATH 11205: Machine Learning in Python 2024-2025

Project 2 Description

We will be using a subset of the data collected by [Freddie Mac](#), called the Single Family Loan-Level Dataset. A simplified version of the data will be used for this project, provided in the file `freddiemac.csv`, after some initial cleaning steps (described below). This smaller data set combines and focuses only on the years 2017-2020. The dataset includes loans that have been paid off, defaulted, or are still active (by the performance cutoff date). Your goal is to build a model to predict if individuals will default on their loans. The data is highly imbalanced, which you should carefully deal with in your analysis.

Assignment Goal

For the purpose of the project, consider yourself a **Data Scientist Consultant** who has been hired by Freddie Mac to analyse loan-level credit performance data on fully amortizing fixed-rate Single-Family mortgages purchased or guaranteed by the company. This data has been made available with goal of helping investors build more accurate credit performance models in support of ongoing risk sharing initiatives highlighted by regulators, specifically, the Federal Housing Finance Agency. For further details on the data, please see the main webpage for the [Single Family Loan-Level Dataset](#).

Towards this aim, you have been asked to use this data to build a classification model to predict if a client will default. You have been instructed to focus **only on discriminating between prepaid and defaulted mortgages**, as the other loans are still active, and thus, may either default or be paid off in the future. In addition, the company is interested in **identifying important factors** that may have an impact on defaulting; this is useful not only to improve their understanding of mortgage defaults, but also to be able explain their decisions to clients and regulatory bodies. In summary, you need to develop a **well-tuned and validated** classification model for **default** as the binary outcome of interest and **interpret or explain** the model's predictions. Your model may use as few or as many of the provided features, transforming and manipulating these features in any way that you see fit, and you may use any additional external sources to create new features. Lastly, you should use your model to inform the company of any **active loans** that may be at risk of default in the future.

You should start from an **interpretable baseline** model of your choice, including as few or as many of the provided variables. At this point, we have covered a number of models in lectures and workshops, and you may explore a variety of different feature engineering and modelling approaches for this particular task. However, your ultimate goal is to select and deliver a **single final model**. Thus, your report should focus on describing and motivating your final model choice, along with a comparison against the baseline model. It is important that any interpretations and conclusions you draw from your model are well supported and sound and that you understand limitations of the model and the data.

Working as a team

This project may be completed by a team of up to 4 students (minimum of 1 student). Feel free to create your own team during workshop hours, building on the pairs for the workshops. Since we are not assigning teams, if you are a team that is looking for more members or someone looking for a team please use the pinned post on Piazza to find each other. You are strongly encouraged to work in teams, as you can learn a lot from discussing together, but you may choose to work individually if preferred. The marking scheme is the same, regardless of group size.

After the assignment is completed, we will distribute a brief peer evaluation survey. Completion of the survey is optional, but if you feel that some members contributed significantly less, this provides an opportunity for feedback and for such members to potentially have their overall mark penalized. This will only be done in extreme cases, after discussion with all team members.

Dataset Details

These are the available variables given in `freddiemac.csv`:

- **fico** - CREDIT SCORE: a number, prepared by third parties, summarizing the borrower's creditworthiness, which may be indicative of the likelihood that the borrower will timely repay future obligations. Generally, the credit score disclosed is the score known at the time of acquisition and is the score used to originate the mortgage. Numeric with values between 300–850 and 9999 for not available (credit scores < 300 or > 850 are shown as **not available**).
- ~~dt_first_pi~~ - ~~FIRST PAYMENT DATE: the date of the first scheduled mortgage payment due under the terms of the mortgage note. Format YYYYMM.~~
- **flag_fthb** - FIRST TIME HOMEBUYER FLAG: indicates whether the Borrower, or one of a group of Borrowers, is an individual who (1) is purchasing the mortgaged property, (2) will reside in the mortgaged property as a primary residence, and (3) had no ownership interest (sole or joint) in a residential property during the three-year period preceding the date of the purchase of the mortgaged property. With certain limited exceptions, a displaced homemaker or single parent may also be considered a First-Time Homebuyer if the individual had no ownership interest in a residential property during the preceding three-year period other than an ownership interest in the marital residence with a spouse. Format: Y=Yes, N=No, 9 = Not Available or Not Applicable.
- ~~dt_matr~~ - ~~MATURITY DATE: the month in which the final monthly payment on the mortgage is scheduled to be made as stated on the original mortgage note. Format YYYYMM.~~
- **cd_msa** - ~~METROPOLITAN STATISTICAL AREA (MSA) OR METROPOLITAN DIVISION: code, with null indicating that the area in which the mortgaged property is located is~~ a) neither an MSA nor a Metropolitan Division, or b) unknown.
- **mi_pct** - MORTGAGE INSURANCE PERCENTAGE (MI %): the percentage of loss coverage on the loan, at the time of Freddie Mac's purchase of the mortgage loan that a mortgage insurer is providing to cover losses incurred as a result of a default on the loan. Amounts of mortgage insurance reported by Sellers that are less than 1% or greater than 55% will be disclosed as "Not Available," which will be indicated 999. No MI will be indicated by zero.
- **cnt_units** - NUMBER OF UNITS: denotes whether the mortgage is a one-, two-, three-, or four-unit property, with 99 indicating Not Available.
- **occpy_sts** - Denotes whether the mortgage type is owner occupied (P), second home (S), or

- investment property (I), or not available (9).
- **cltv** - ORIGINAL COMBINED LOAN-TO-VALUE (CLTV): with 999 indicating not available.
- **dti** - ORIGINAL DEBT-TO-INCOME (DTI) RATIO: disclosure of the debt to income ratio is based on (1) the sum of the borrower's monthly debt payments, including monthly housing expenses that incorporate the mortgage payment the borrower is making at the time of the delivery of the mortgage loan to Freddie Mac, divided by (2) the total monthly income used to underwrite the loan as of the date of the origination of the such loan. Ratios greater than 65% are indicated that data is Not Available (999).
- **orig_upb** - The unpaid balance (UPB) of the mortgage on the note date (rounded to the nearest \$1,000).
- **ltv** - ORIGINAL LOAN-TO-VALUE (LTV). range prior to 2018Q1: 6% - 105% and range post to 2018Q2: 1% - 998%, with 999 = Not Available
- **int_rt** - The interest rate of the loan as stated on the note at the time the loan was originated.
- **channel** - indicates whether a Broker or Correspondent, originated or was involved in the origination of the mortgage loan. If a Third Party Origination is applicable, but the Seller does not specify Broker or Correspondent, the disclosure will indicate "TPO Not Specified". Similarly, if neither Third Party Origination nor Retail designations are available, the disclosure will indicate "TPO Not Specified." If a Broker, Correspondent or Third Party Origination disclosure is not applicable, the mortgage loan will be designated as Retail. Values: R = Retail, B = Broker, C = Correspondent, T = TPO Not Specified, 9 = Not Available.
- **ppmt_pnlty** - PREPAYMENT PENALTY MORTGAGE (PPM) FLAG: denotes whether the mortgage is a PPM. A PPM is a mortgage with respect to which the borrower is, or at any time has been, obligated to pay a penalty in the event of certain repayments of principal.
- **prod_type** - Denotes that the product is a fixed-rate mortgage or adjustable-rate mortgage.
- **st** - A two-letter abbreviation indicating the state or territory within which the property securing the mortgage is located.
- **prop_type** - Denotes whether the property type secured by the mortgage is a condominium (CO), planned unit development (PU), cooperative share (CP), manufactured home (MH), or Single-Family home (SF). If the Property Type is Not Available, this will be indicated by 99.
- **zipcode** - The postal code for the location of the mortgaged property. Format ###00, where ### represents the first three digits of the 5 digit postal code and 00 = Unknown.
- **id_loan** - Unique identifier assigned to each loan
- **loan_purpose** - Indicates whether the mortgage loan is a Cash-out Refinance mortgage (C), No Cash-out Refinance mortgage (N), Refinance mortgage not specified (R), or a Purchase mortgage (P), with 9 =Not Available.
- **orig_loan_term** - ORIGINAL LOAN TERM: the number of scheduled monthly payments of the mortgage based on the First Payment Date and Maturity Date.
- **cnt_borr** - The number of Borrower(s) who are obligated to repay the mortgage note secured by the mortgaged property. Disclosure denotes only whether there is one borrower (1), or more than one borrower associated with the mortgage note (2).
- **seller_name** - SELLER NAME: the entity acting in its capacity as a seller of mortgages to Freddie Mac at the time of acquisition.
- **servicer_name** - SERVICER NAME: the entity acting in its capacity as the servicer of mortgages to Freddie Mac as of the last period for which loan activity is reported in the Dataset.

- **flag_sc** - SUPER CONFORMING FLAG: For mortgages that exceed conforming loan limits with origination dates on or after 10/1/2008 and were delivered to Freddie Mac on or after 1/1/2009. Values: Y=Yes and N=NaN.
- **id_loan_rr** - ~~PRE-RELIEF REFINANCE LOAN SEQUENCE NUMBER - Populated only for loans where the Relief Refinance Indicator is set to Y.~~
- **program_ind** - The indicator that identifies if a loan participates in any of the programs: H = Home Possible, F = HFA Advantage, R= Refi Possible, 9 = Not Available or Not Applicable.
- **rr_ind** - Indicator that identifies whether the loan is part of Freddie Mac's Relief Refinance Program. Loans which are both a Relief Refinance and have an Original Loan-to-Value above 80 are HARP loans. Values: Y=Yes and N=NaN.
- **property_val** - The indicator denoting which method was used to obtain a property appraisal, if any. Values: 1 = ACE Loans, 2 = Full Appraisal, 3 = Other Appraisals (Desktop, driveby, external, AVM), 4=ACE+PDR, 9 = Not Available.
- ~~io_ind - The indicator denoting whether the loan only requires interest payments for a specified period beginning with the first payment date.~~
- ~~mi_cancel_ind - The indicator denoting if the mortgage insurance has been reported as canceled after the time of Freddie Mac's purchase of the mortgage loan. If a loan did not have mortgage insurance at the time of Freddie Mac's purchase of the mortgage loan, then this field will be disclosed as Not Applicable. Values: Y = Canceled, N = Not Canceled, 7 = Not Applicable, 9 = Not Disclosed.~~
- **loan_status** - whether person defaulted on the loan, paid of the loan, or if loan is still active.

For further details, please see the **Data User Guide** provided in project materials (in particular **pgs. 5-10 for a summary table of the data**). I have created the target variable `loan_status` for you, which utilizes the monthly performance data to identify if loans have defaulted, been paid off, or are still active (due to the size of this monthly data, it is not included in the project materials).

You may choose to utilise additional data sources, for example, when encoding categorical features, such as state or zipcode, if you feel that it is useful. In this case, please describe and reference any additional sources.

The data provided for this project is subject to data agreements and waivers that I have signed on your behalf. Thus, **the data must NOT be shared publicly** (e.g. if your team is using GitHub, keep the repo private and do NOT share the data with any GenAI tools).

Required Structure

A Jupyter notebook template called `project2.ipynb` has been provided. It includes the required sections along with brief instructions on what should be included in each section. Your completed assignment must follow this structure - **you should not add or remove any of these sections, but you may add extra subsections to help organize the report**. Please remove the instructions for each section in the final document.

All of your work must be contained in the `project2.ipynb` notebook, we will only mark what is included in this file (both the write-up and relevant coding). You may work on the notebook in whichever environment you prefer (noteable, locally, colab, codespaces,...).

There is an **upper limit of 30 pages** including all code and output. Your notebook must

include all of your work, but make sure that you are only retaining required components, e.g. remove unused code and figures (if a figure is not explicitly discussed in the text it should not be in the final document). Overall, your project will be partially assessed on your organization / presentation of the document - it should be as polished and streamlined as possible. **Try to be as concise as possible while creating your write-up. We highly recommend that you check the appearance of your rendered PDF before submitting, as its appearance can differ significantly from the notebook.**

Please submit your final PDF of project report (generated from a Jupyter notebook) to the Project assignment on Gradescope. Please ensure that you **tag all groups members** on Gradescope, and also add all group member names either in the notebook metadata or in additional markdown cell block at the beginning of the file.

Getting Help

- **Week 11 Workshop:** There will be only an unmarked notebook on explainable AI tools during Week 11. We will focus on answering any project related questions.
- **Piazza:** This forum will be used as the central location for all course related discussions and questions, and should be used over emailing course staff directly. The course lecturers will monitor and respond to questions, but feel free to provide some constructive responses to peer's questions.

Generative AI Policy

Please refer to the School of Mathematics [Generative AI Policy](#). Generative AI tools may be used for project, but always as your helper or co-pilot and used responsibly, and NOT your driver! For example, GenAI may be used for generating explanations for error messages and debugging, providing hints or suggestions to improve code, enhancing visualizations and the quality of the report. The data should absolutely NOT be copied and shared directly with GenAI tools, as this is in breach of legal consent for this data. You must include a statement on how GenAI was used. **If the project is suspected to be heavily written by GenAI, the students may be subject to an oral presentation.**

Further References

We have provided additional resources in the project materials.

For further information on the dataset and variables, see:

- *Single Family Loan-Level Dataset:*
<https://www.freddiemac.com/research/datasets/sf-loanlevel-dataset>

For an example of a published analysis of a subset of this data, see:

- Bhattacharya et al (2019) *A Bayesian approach to modeling mortgage default and prepayment*. European Journal of Operational Research, 274: 1112-1124.
<https://www.sciencedirect.com/science/article/pii/S0377221718309159>